Research Notes 1

# AMFU-Net: A Multi-Scale Attention Network for Lesion Segmentation in Fused Breast Ultrasound Imaging

data augmentation, methodology, dataset comparison (BUSI), larger datasets, and data access.

# How the method works

## The method has two stages:

- Stage 1 – Fusion: combine two ultrasound images into one better image.
- Stage 2 – Segmentation: use a neural network (AMFU-Net) to outline the lesion in that fused image.

## The data:

- Images come from a USCT machine that scans the breast using a ring of sensors around it.
- It captures two images at the same time: a reflection image (shows shape and structure) and a transmission image (shows tissue properties such as sound speed).
- Doctors labelled the lesions by hand to make “masks” (the correct answer). There are 215 images, each 256 × 256 pixels.
- The images were split into 80% for training, 10% for checking (validation), and 10% for the final test.

# Step by step the full process

## Stage 1: Make the fused image

- 1. Convert the transmission image into HIS format and take its brightness (I) part.
- 2. Use the Wavelet Transform (DWT) to split each image into a blurry part (structure) and sharp parts (edges).
- 3. Combine the blurry parts using PSO , an algorithm that automatically finds the best mix.
- 4. Combine the sharp parts by keeping the strongest pixels.
- 5. Sharpen and clean the result, then turn it back into a normal image.

## Stage 2: Find the lesion + test

- 6. Feed the fused image into AMFU-Net. This is a U-Net with two upgrades: DAG attention gates (in the skip connections) and an MFEF module (at the centre/bottleneck) to read features at many sizes.
- 7. The network outputs a mask showing where the lesion is.
- 8. Train with the Adam optimiser for 200 rounds; measure accuracy with Dice, Jaccard, 95HD and ASD.
- 9. Result: 91.42% Dice , better than U-Net and five other networks.

# Did data augmentation increase the size of the data?

**it did not increase the real dataset.**

- They started with 215 images (from 210 patients). This number never changed.
- Augmentation only made altered copies of the training images: flipping, rotating, scaling, and shifting them.
- This helps the model see more variety and stops it from memorising, but it adds no new patients and no new real information.
- The copies are not counted as extra data. The whole study is still based on the same 215 images.
- The authors themselves say the dataset is small and comes from only one type of machine at one place.

QUESTION 3

# Their dataset vs the BUSI (Kaggle) dataset

**Most important: BUSI cannot be used for the fusion part, because it has no transmission image.**

| Point | Their data (USCT) | BUSI (Kaggle) |
|---|---|---|
| Machine used | Special ring USCT scanner | Normal handheld ultrasound probe |
| Images per scan | Two: reflection + transmission | One image only |
| Can do the fusion step? | Yes | No |
| Size | 215 images, 210 patients | 780 images, 600 patients |
| Image types | Lesion masks only (all have lesions) | Normal, benign, malignant |
| Colour | Colour | Greyscale |
| Image resolution | 256 × 256 (fixed) | About 500 × 500 (varies) |
| Country | China | Egypt |
| Public? | No (private) | Yes (on Kaggle) |

**Summary**

**BUSI (Breast Ultrasound Images)**

Each **case** represents **one lesion from one patient examination**.
A **single handheld ultrasound probe** both sends sound waves and receives the echoes that reflect back, producing **one grayscale reflection image**.
The accompanying **mask image** is **not another scan**; it is a manually created annotation outlining the lesion.
Since the probe is only on **one side of the breast**, it can only capture **reflected sound waves**. Sound that passes through the breast is not detected, so **BUSI contains only reflection images**.

**Chinese USCT (Ultrasound Computed Tomography) System**

The patient lies face-down with the breast suspended in a **water tank** surrounded by a **360° ring of 512 transducers**.
During a single scan:

Sensors near the transmitting element capture **reflected echoes**, creating a **reflection image**.
Sensors on the opposite side capture **transmitted sound waves**, creating a **transmission image** that provides information about tissue properties such as sound speed, density, and stiffness.

As a result, **one scanning session produces two complementary images of the same lesion**: a reflection image and a transmission image.

QUESTION 4

# Alternative Datasets

| Dataset | Images | Patients | Notes |
|---|---|---|---|
| BUS-BRA | 1,875 | 1,064 | The largest public one. Has lesion masks and extra labels (Brazil). |
| BUS-UCLM | 683 | 38 | Has masks for benign and malignant lesions (Spain). |
| BUSI | 780 | 600 | Statistically categorized into three distinct clinical profiles: Normal, Benign, and Malignant (Egypt). |
| BrEaST | 256 | 256 | Collected with four different machines (Poland). |
| UDIAT | 163 | — | Small and older, but widely used for comparison (Spain). |

Important: all of these use normal handheld ultrasound (one image per scan). There is no public USCT dataset with reflection + transmission images. So none of them can fully copy this study’s fusion method ,they can only be used for the segmentation part.

QUESTION 5

# Why we cannot access their dataset

**The authors wrote that they do not have permission to share the data. The main reasons:**

- Patient privacy: patients agreed to join the study, but not necessarily to having their images shared publicly.
- Ethics approval: the hospital ethics board (IRB) approved the data only for this study, not for public release.
- Data laws: the data comes from two hospitals in China, which has strict rules about sending medical/patient data out of the country.
- Special machine: the images come from a custom scanner, so the raw data is in a special format that is hard to share.

**Interesting points:**

- Sharing would need sign-off from two separate hospitals, which makes it even harder.
- The authors admit this single-hospital, single-machine dataset is a weakness of their study. Why are going with this paper???